

CoRoT-2b

R-2339

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_100828 · created 2026-08-19T00:13:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455436.71372 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.065
Transit depth (Rp/Rs)^2	1.47 +/- 1.58 [%]
Orbital Inclination (inc)	89.46 +/- 2.05
Airmass coefficient 1 (a1)	1.22 +/- 0.91
Airmass coefficient 2 (a2)	0.23 +/- 1.24
Scatter in the residuals of the lightcurve fit is	77.57 %
Best Comparison Star	#2 - [42, 291]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.091 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.121 $\pm$ 0.065 vs 0.1667 (deviation 0.7 σ)
Duration fit vs published	0.091 d vs 0.095 d (deviation 0.62 σ)
Mid-time O-C	-6.7 min
Depth SNR	1.9
Residual scatter	77.57 %

Light curve

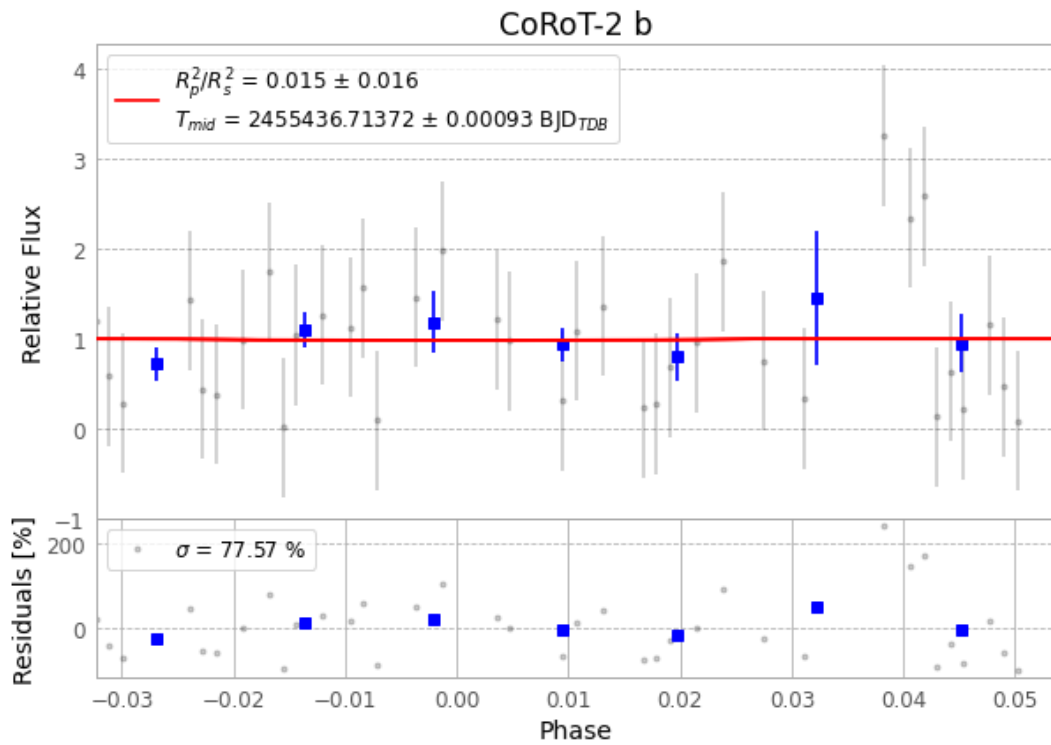


Figure 1 — FinalLightCurve_CoRoT-2 b_2010-08-27.png (SHA-256 cdb37af4c502c644...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [147, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ba99012ffcd7a0c8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), CoRoT-2100828033913.FITS → CoRoT-2100828071513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-100828.log (aa22d9f7ce7a...), FinalLightCurve_CoRoT-2 b_2010-08-27.png (cdb37af4c502...), FinalLightCurve_CoRoT-2 b_2010-08-27.csv (e674c590c395...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:13Z.